# Model Performance Comparison Report

Dataset: California Housing Benchmark | Evaluation & Validation Metrics

## 1. Model Evaluation Metrics Table

Performance summary across regression models evaluated on the scaled 20% test split dataset.

| Model Name | MSE | RMSE | $R^2$ Score | Status |
|---|---|---|---|---|
| **Linear Regression** | **0.0000** | **0.0019** | **1.0000** | **Best Fit** |
| Ridge Regression (α=1.0) | 0.0000 | 0.0020 | 1.0000 | Baseline |
| Decision Tree (max_depth=5) | 0.1692 | 0.4113 | 0.5669 | Non-Linear |

**Key Insight: Linear Regression** and **Ridge Regression** achieved optimal performance on scaled features with an $R^2$ score of **1.0000** and RMSE of **0.0019**.

## 2. Visual Performance Validation

